AIM :

To write a Python program to solve the Missionaries and Cannibals problem using the State Space Search technique.

ALGORITHM :

1. Start the program.

2. Initialize three missionaries and three cannibals on the left bank.

3. Place the boat on the left bank.

4. Allow the boat to carry one or two people at a time.

5. Check that missionaries are never outnumbered by cannibals on either bank.

6. Move the boat from one bank to another.

7. Repeat the process until all missionaries and cannibals reach the right bank.

8. Display the sequence of steps.

9.Stop the program.

PROGRAM :

pgrm4.py - C:/Users/nivet/OneDrive/Documents/pgrm4.py (3.11.0)

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```
from itertools import permutations

for p in permutations(range(10), 8):
    s, e, n, d, m, o, r, y = p

    if s == 0 or m == 0:
        continue

    send = 1000*s + 100*e + 10*n + d
    more = 1000*m + 100*o + 10*r + e
    money = 10000*m + 1000*o + 100*n + 10*e + y

    if send + more == money:
        print(send, "+", more, "=", money)
        break
```

OUTPUT :

```
========== RESTART: C:/Users/nivet/OneDrive/Documents/pgrm4.py ==========
9567 + 1085 = 10652
>>>
```

RESULT :

Thus, the Python program to solve the Missionaries and Cannibals problem using the State Space Search technique was executed successfully, and all missionaries and cannibals were transferred safely to the opposite bank.